Genome QC · assessment to exploration

Inputs: genome FASTAs + reference databases

Input · Compute · Data · Output

Preprocessing 1

Genome FASTAs → Stage / contig variants (Genome QC, SeqKit) → Assembly statistics (SeqKit) → Nonempty variants

Quality assessment 2

Nonempty variants → Marker assessment (CheckM lineage_wf) → Calculate Qscore (Genome QC) → Quality metrics

Taxonomic context 3

Assessed variants → Classify genomes (GTDB-Tk, R226) → Extract lineage / domain (Genome QC) → Taxonomy table

rRNA evidence 4

Variants + domain → Predict rRNA genes (barrnap) → Self-BLAST / consistency (BLAST+, Genome QC) → RNA consistency

Representative 5

Variant evidence → Choose variant (Qscore, N50, length) → Predict tRNAs (tRNAscan-SE) / Assess chimerism (GUNC) → tRNA + GUNC

Final selection 6

All representatives → Join metrics + status (Genome QC) → Apply requested bounds (Default: unrestricted) → Master + selected

Genome atlas 7

Complete master → Annotate quality metrics (pandas, NumPy) → Summarize / plot (Matplotlib, seaborn) → Atlas tables + plots

Explorer 8

Tables + atlas → Index linked results (Genome QC) → Filter / document subsets (Genome QC Explorer) → CSV + JSON record